

Exam 2019 - sample solution

Does conference call participation
by institutional investors affect the market?

This sample solution is one of many possible solutions. The grade for the exam will not be based on how closely you follow a particular suggested solution but rather on the overall quality of your empirical analysis.

Abstract

We investigate whether conference call participation by institutional investors affects the market. We find that institutional investors might be distracted if there are many other conference calls or merger activity on the same day or within the same industry. Thus, we use the number of merger announcements and number of calls in the same day and industry as instruments for participation by institutional investors. We find a weak negative effect on the abnormal returns and a strong positive effect of participation by institutional investors on abnormal trading volume.

*This suggested solution is for the Fall 2019 exam created by Maximilian Rohrer.

1 Introduction

In this study, we investigate whether conference call participation by institutional investors affects the market. The assignment description provides some theoretical evidence on why and how this relationship might be present.

First, we find that roughly 16% of the firms experience conference call participation by institutional investors. We document that the participation by institutional investors in a call depends on several variables such as number of participants, market value and price of the firm prior to the conference call, and sales-to-assets ratio. We construct a matched sample to address the issue that firms that experience institutional participation in a call are different from firms that did not experience institutional participation.

Second, we show that institutional participation in a call depends on number of merger announcements and number of calls in the same day and industry for full and matched samples alike. We find that those variables are significantly negatively related to institutional investors' participation. This finding confirms the assignment description that institutional investors might be distracted if there are many other conference calls or merger activity on the same day or within the same industry.

Finally, we use those variables as instruments for participation by institutional investors and investigate the effect of conference call participation by institutional investors on the market. We find that institutional participation is negatively related to abnormal returns for the full sample, but not for the matched sample. We find a strong positive effect of participation by institutional investors on abnormal trading volume for full and matched samples alike.

2 Data

We use data on conference calls which includes information on participants and timing of participation.¹ Quarterly accounting data come from Compustat. Furthermore, we use data on

¹Data is provided and detailed description can be found in the assignment. Some variables have to be calculated, we refer to the R-code for detailed variable construction. Selected variables have been winsorized on a 99% level.

Fama-French 49 industries and on mergers that occurred during our sample period.

Panel A of Table 1 shows the summary statistics for firms with institutional participation in the conference calls, while Panel B shows the summary statistics for firms without institutional participation in the conference calls. Roughly 16% of firms experience conference call participation by institutional investors. We refer to the next section for discussion on the determinants of the institutional participation in the conference calls.

The abnormal return is on average negative if institutional investors participate and positive if they do not. Abnormal trading volume is on average only slightly lower if institutions participate in a call than if they do not participate in a call. This constitutes preliminary evidence that participation of institutional investors in a call has an effect on a stock market. However, summary statistic is not enough to draw credible conclusion on the causal effect of the institutional investors' participation on the stock market. We will proceed with the formal regression analysis in the next section.

Since only 16% of the firms experience institutional investors' participation in a call, we construct a matched sample from the firms that do not experience institutional investors' participation. In particular, we use propensity score matching based on the number of participants (log), market value in the end of last month (log), price end of last month (log), sales-to-assets ratio, SUE, and year. Further analysis is conducted on both full and matched samples.

3 Empirical analysis

First, we identify observable factors which can help us explain conference call participation by institutional investors. We model probability of institutional investors' participation using linear probability and logit models. We regress a dummy for whether institutional investors participate in a conference call on the the number of participants (log), market value in the end of last month (log), price end of last month (log), sales-to-assets ratio, SUE, and year fixed effects. We also add number of merger announcements and number of calls in the same day and industry as variables

explaining the decision of institutional investors to participate in a call.

Table 2 presents the results of the above mentioned analysis. Columns (1) to (3) present coefficients from the linear probability regression while columns (4) to (6) present the marginal effects from the logit regression. Furthermore, in columns (3) and (6), the matched sample is used while in all other columns, the full sample is used.

The regression results show that for the unmatched sample, number of participants, market value and price of the firm prior to the conference call, and sales-to-assets ratio are important determinants of institutional participation. We note that for the matched sample these variables do not determine institutional participation any more which is a signal of a quality of our matching procedure.

Furthermore, number of merger announcements and number of calls in the same day and industry are significantly negatively related to institutional investor participation. In particular, for the matched sample, increase in the number of merger announcements (number of other calls in the same day and industry) by one leads to a decrease in probability of participation of 0.9% (0.4%) for both linear probability and logit models and this decrease is significant at the 1% level. This finding confirms the assignment description that institutional investors might be distracted if there are many other conference calls or merger activity on the same day or within the same industry.

Next, we investigate the link between institutional investors conference call participation and the market reaction. We use two proxies for market reaction: the abnormal return and the natural logarithm of abnormal trading volume.

We note that there might be other explanations for potential market reaction that might be not directly linked to institutional investors conference call participation (potential unobservable factors). To address endogeneity concerns, we use the instrumental variable approach. The analysis above shows that the number of merger announcements and number of calls in the same day and industry are significantly negatively related to institutional investor participation. We test for instrument relevance by means of an F -test. We show that F -statistics has a value of

43.36 and 19.86 for the full and matched samples, respectively. These values are greater than 10 and thus, the instruments are relevant. In addition, these variables are exogenous with respect to a particular firm in question. Therefore, we use these variables as instruments for institutional investors' participation.

Table 3 presents the results of the instrumental variable analysis. Columns (1) and (3) present the results of the 2SLS regression for the full sample while columns (2) and (4) present the results for the matched sample. We note that matching procedure enhances reliability of our results and therefore, we consider the estimates from the matched sample to be more precise. Institutional investor participation is negatively related to the abnormal returns (significant at the 10% level for the full sample) and positively related to abnormal volume (significant at the 1% level for both samples).

Overall, while the evidence for abnormal returns is not straightforward, the 2SLS estimation results show that institutional investor participation in conference calls increases abnormal trading volume.

4 Conclusion

We investigate whether conference call participation by institutional investors affects the market. We find that the number of merger announcements and number of calls in the same day and industry are significantly negatively related to institutional investor participation which confirms that institutional investors might be distracted if there are many other conference calls or merger activity on the same day or within the same industry. We use those variables as instruments for participation by institutional investors and investigate the effect of conference call participation by institutional investors on the market. From our analysis, we cannot draw strong conclusions on the effect on the abnormal returns, however, we find a strong positive effect of participation by institutional investors on abnormal trading volume.

Table 1: **Summary statistics of the most important variables used in this study**

This table shows the summary statistics for the variables used in this study. Panel A (B) shows the results for firms with institutional investors (not) participating in conference calls. Detailed variable definitions can be found in the R-code or the assignment description. Selected variables have been winsorized on a 99% level.

Panel A: Firms with institutional investors participating in conference calls

	mean	sd	min	max	nr.obs
Institutional participation	1.000	0.000	1.000	1.000	8,225.000
Abnormal return	-0.058	9.951	-69.876	88.085	8,225.000
Abnormal volume	0.591	0.653	-1.349	2.591	8,225.000
Participants	6.430	3.626	1.000	28.000	8,225.000
MV	6.369	1.873	0.713	12.914	8,225.000
Price	2.580	1.128	-2.526	5.505	8,225.000
Sales-to-assets ratio	0.254	0.194	0.000	1.085	8,225.000
SUE	-0.010	1.002	-5.000	5.000	8,225.000
Mergers	0.942	1.410	0.000	9.000	8,225.000
Calls on the same day and industry	4.131	4.998	0.000	33.000	8,225.000

Panel B: Firms with institutional investors not participating in conference calls

	mean	sd	min	max	nr.obs
Institutional participation	0.000	0.000	0.000	0.000	40,888.000
Abnormal return	0.094	9.032	-92.364	147.610	40,888.000
Abnormal volume	0.652	0.585	-1.349	2.591	40,888.000
Participants	7.252	3.974	1.000	27.000	40,888.000
MV	7.386	1.869	0.732	13.481	40,888.000
Price	3.021	1.110	-2.928	6.728	40,888.000
Sales-to-assets ratio	0.240	0.189	0.000	1.085	40,888.000
SUE	0.026	0.946	-5.000	5.000	40,888.000
Mergers	0.678	1.267	0.000	11.000	40,888.000
Calls on the same day and industry	4.875	5.550	0.000	33.000	40,888.000

Table 2: **Factors explaining conference call participation by institutional investors**

This table shows factors that explain institutional participation in conference calls. We regress a dummy for whether institutional investors participate in a conference call on the the number of participants (log), market value in the end of last month (log), price end of last month (log), sales-to-assets ratio, SUE, and year fixed effects. We also add number of merger announcements and number of calls in the same day and industry as variables explaining the decision of institutional investors to participate in a call. Columns (1) to (3) present the results of the linear probability regression while columns (4) to (6) present the results of the logit regression. Furthermore, in columns (3) and (6), the matched sample is used while in all other columns, the full sample is used. Detailed variables' definitions can be found in the R-code or the assignment description. Selected variables have been winsorized on a 99% level. Standard errors are robust.

	<i>Dependent variable:</i>					
	Institutional participation					
	(1)	(2)	(3)	(4)	(5)	(6)
log(Participants)	0.060*** t = 19.341	0.059*** t = 18.953	0.005 t = 0.597	0.063*** t = 18.308	0.061*** t = 17.987	0.005 t = 0.598
log(MV)	-0.056*** t = -37.238	-0.056*** t = -37.338	-0.006 t = -1.585	-0.064*** t = -33.386	-0.064*** t = -33.233	-0.006 t = -1.591
log(Price)	0.009*** t = 4.254	0.009*** t = 3.977	0.004 t = 0.747	0.018*** t = 7.629	0.017*** t = 7.305	0.004 t = 0.748
Sales-to-assets ratio	-0.012 t = -1.333	-0.027*** t = -3.035	-0.013 t = -0.636	-0.010 t = -1.188	-0.023*** t = -2.632	-0.013 t = -0.637
SUE	-0.0003 t = -0.195	-0.0001 t = -0.073	0.001 t = 0.282	0.0003 t = 0.180	0.001 t = 0.329	0.001 t = 0.281
Mergers		-0.005*** t = -2.750	-0.009*** t = -2.618		-0.004** t = -2.485	-0.009*** t = -2.621
Calls		-0.003*** t = -9.279	-0.004*** t = -5.675		-0.002*** t = -7.359	-0.004*** t = -5.620
Constant	0.551*** t = 37.854	0.577*** t = 37.543	0.545*** t = 17.608			
Sample	full	full	match	full	full	match
Model	OLS	OLS	OLS	logit ME	logit ME	logit ME
Year FE	yes	yes	yes	yes	yes	yes
Observations	49,113	49,113	16,450	49,113	49,113	16,450

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 3: **Institutional investors conference call participation and the market reaction**

This table shows the results of the instrumental variables regression. In particular we study the market reaction as proxied by abnormal return and abnormal trading volume to institutional investors' participation in conference calls. We use number of merger announcements and number of calls in the same day and industry as instruments for institutional participation. All our regressions are estimated with year fixed effects. Columns (1) and (3) present the results of the 2SLS regression for the full sample while columns (2) and (4) present the results for the matched sample. Detailed variable definitions can be found in the R-code or the assignment description. Selected variables have been winsorized on a 99% level. Standard errors are robust.

	<i>Dependent variable:</i>			
	Abnormal return		log(Abnormal volume)	
	(1)	(2)	(3)	(4)
Institutional participation	-5.493* t = -1.956	-1.772 t = -0.562	2.715*** t = 8.117	1.501*** t = 4.761
log(Participants)	0.558*** t = 2.900	0.352** t = 1.984	0.037 t = 1.614	0.182*** t = 10.692
log(MV)	-0.408** t = -2.534	-0.114 t = -1.479	0.090*** t = 4.687	-0.040*** t = -4.769
log(Price)	0.251*** t = 3.565	0.087 t = 0.691	0.058*** t = 7.461	0.077*** t = 6.911
Sales-to-assets ratio	1.159*** t = 4.935	1.466*** t = 3.356	0.436*** t = 15.415	0.408*** t = 10.003
SUE	1.036*** t = 18.520	1.147*** t = 10.493	0.011* t = 1.914	0.018** t = 2.279
Constant	2.610* t = 1.672	0.237 t = 0.143	-1.013*** t = -5.373	-0.366** t = -2.143
Observations	49113	16450	49113	16450
Sample	full	match	full	match
Year FE	yes	yes	yes	yes
Adjusted R ²	0.014	0.014	0.067	0.067

Note:

*p<0.1; **p<0.05; ***p<0.01